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Small Changes, Big Trouble

Demystifying and Parsing License Variants
for Incompatibility Detection in the PyPI Ecosystem



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Motivation: Why License Variants Matter



Open-source licenses are the legal foundation

License variants are licenses that differ from standard SPDX texts — either modified or fully custom.



Even small changes can introduce significant legal differences

Existing tools do not account for license variants, leading to missed incompatibilities and compliance risks.



Non-compliance → ethical, legal, monetary consequences

Real-World Example

usd-core (580K+ downloads)

Appears to be Apache 2.0 but modifies Section 6 (Trademarks), imposing more restrictive conditions.

cuda-python (NVIDIA)

Uses a fully custom "NVIDIA SOFTWARE LICENSE" — prohibits modification and redistribution entirely.

Research Gaps & Our Goals

Gap 1: Knowledge Gap

Little is known about the prevalence, characteristics, and impact of license variants in packaging ecosystems.

Gap 2: Tool Gap

No efficient license parsing or incompatibility detection approaches account for these variants.

Our Contributions



Empirical Study

First comprehensive study of license variants in PyPI



LV-Parser

Efficient diff-based license parsing with LLMs



LV-Compat

Precise incompatibility detection pipeline

Empirical Study Design



Study Subject & Data

Top 8,000 most downloaded PyPI packages

5,990 license files obtained

554,079 packages / 6.19M releases

Full dependency trees constructed



Research Questions

RQ1: Prevalence, distribution, and textual characteristics of license variants?

RQ2: How do license variants impact compatibility in dependency networks?

Methodology

1

Winnowing Algorithm

Compare against SPDX texts

2

Manual Review

Two authors classify ($\kappa=0.87$)

3

Dependency Analysis

Build full dep trees

4

Compatibility Check

Detect incompatibilities

RQ1: Distribution of License Variants

Textual variations are widespread, but only 2% involve substantive modifications.

Type	Count	Percentage
SPDX License	5,869	97.98%
Dual / Multiple License	33	0.55%
Modified SPDX License	18	0.30%
Fully Custom License	69	1.12%

Winnowing Similarity Scores

0.90

Median Score

0.96

95th Percentile

Additional Findings: 144 packages (2.4%) include third-party licenses in their license files. 11 of 18 modified licenses include exception statements. PyTorch's license file spans 8,850+ lines.

RQ2: Impact on License Compatibility

10.7%

Downstream packages
with incompatibilities

9.0%

Incompatible
releases (3,070)

21.8%

Uncertain releases
(7,426 unknown)

Key Findings



1,879 cases (5.5%) involve proprietary prohibitions — free software depending on proprietary upstream



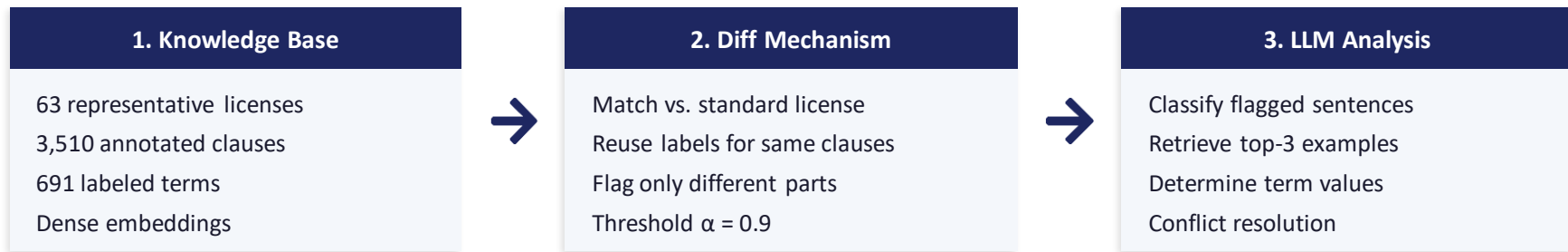
Stricter variants → 21% increase in incompatible downstream releases vs. SPDX originals



Relaxing exceptions → 76% reduction in incompatibilities vs. SPDX originals

LV-Parser: Design Overview

Key Insight: Only a small fraction of variants have substantive changes → avoid redundant analysis of standard provisions.



Design Advantages

Efficient

Skip analysis of identical standard provisions — only parse differences

Accurate

Knowledge base + few-shot examples provide rich context for LLMs

Robust

Works across different LLMs (GPT-4.1, Claude, DeepSeek, Qwen)

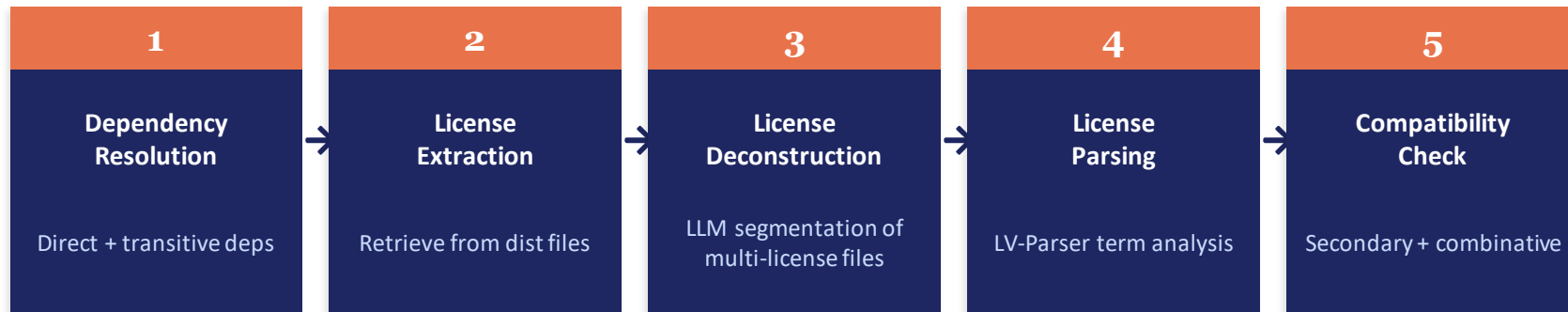
LV-Parser Evaluation

Evaluation Dataset: 74 licenses (20 standard SPDX + 54 modified/custom) · Inter-annotator $\kappa = 0.86$

Approach	Avg. Accuracy	LLM Queries/License	Cost Reduction
LV-Parser (GPT-4.1)	0.936	42	~30%
Baseline (GPT-4.1)	0.894	59	-
LV-Parser (Claude Sonnet 4)	0.901	-	-
LV-Parser (DeepSeek-v3)	0.887	-	-
LV-Parser (Qwen3-14B)	0.895	-	-

Takeaway: LV-Parser outperforms the baseline across accuracy (+4.2%) and efficiency (~30% fewer LLM calls). Even small models (Qwen3-14B) achieve 0.895 accuracy — the method works, not just the model.

LV-Compat: Incompatibility Detection Pipeline



Compatibility Framework

Secondary Compatibility

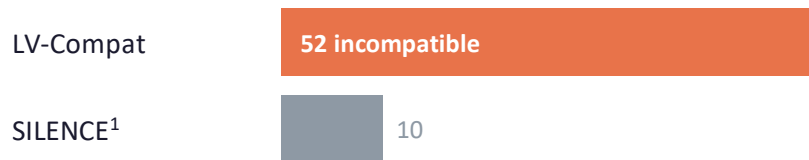
All parts of derivative work can be re-licensed under the downstream license. Requires matching or stricter copyleft + satisfying all terms.

Combinative Compatibility

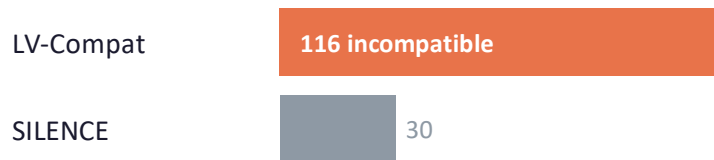
Components retain original licenses while the project adopts a different license (e.g., MIT project using LGPL dependency).

LV-Compat vs. SILENCE: Results

Package-Level Detection (out of 75)



Dependency-Level Detection



	Incompat. Packages	Unknown Packages	Incompat. Deps	Unknown Deps	3rd-Party Incompat.
LV-Compat	52	7	116	52	238
SILENCE	10	65	30	598	N/A

5.2×

More incompatible
packages detected

0.98

Precision
(51/52 correct)

92% ↓

Fewer unknown
dependencies

[1] Xu W, He H, Gao K, et al. Understanding and remediating open-source license incompatibilities in the pypi ecosystem[C]//2023 38th IEEE/ACM International Conference on Automated Software Engineering (ASE). IEEE, 2023: 178-190.

Discussion: Insights & Challenges

Challenging Cases

Copyright classification

Ambiguous vs. explicit grants — boundary is often unclear

Usage limitation (open-ended)

Difficult to enumerate all restrictions precisely

Legal ambiguity

Even human annotators disagree on patent clause interpretation

Limitations

Focus on top PyPI packages

May not reflect full ecosystem diversity

Manual annotation subjectivity

Legal language is inherently ambiguous

Package-level only

Does not consider in-source-code license info

Root Cause of "Unknown" Cases

All 7 unknown packages in LV-Compat stem from incomplete license information upstream (e.g., "BSD" without specifying version). This is an inherent ecosystem data quality issue, not a tool limitation.

Conclusion



First empirical study on license variants in PyPI

Textual variations are widespread; 2% are substantive; 10.7% downstream packages are incompatible



LV-Parser: Efficient diff-based license parsing

0.936 accuracy, ~30% cost reduction, robust across multiple LLMs



LV-Compat: Precise incompatibility detection

5.2× more incompatible packages detected with 0.98 precision

Data & Code: <https://figshare.com/s/4fbaedbeb120d1940f12>

Thank You!

Questions?

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